DAY SHIFT
AGE-RESPECTFUL UPPER-LIMB REHABILITATION GAMES

CS 435 – HCI Group Course Project

Assignment 3: High-Fidelity Prototyping and User Evaluation

Group 1 · Demo video:
https://drive.google.com/file/d/1J38DBpJ5eY5m0iKNOVhc_1249CBgIX_T/view?usp=sharing
Repository: https://github.com/Jain-Laksh/HCI-Upper-Limb-Rehabilitation-Gamified-Platform
Website: https://jain-laksh.github.io/HCI-Upper-Limb-Rehabilitation-Gamified-Platform

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Abstract

Day Shift is a home tele-rehabilitation game for older adults who are working to recover upper-limb function. Instead of counting repetitions, the player spends a working day as a **house painter**. Each exercise is drawn as a strip of masking tape on a wall, the player's hand acts as the brush, and paint only goes on while the movement stays on the tape. Our high-fidelity prototype is a web app that runs entirely in the browser with an ordinary laptop webcam. Exercise paths are stored relative to the player's own body, so one reference movement scales to people of different sizes. We recover movement toward the camera from calibrated bone lengths, and a therapist can create a new exercise from a phone video. The interface was designed around the perceptual and motor changes that come with ageing, with large high-contrast text, dwell selection with the free hand, spoken instructions with captions, voice commands and Hindi. We tested the prototype with nine participants. Usability was good (mean SUS 73.3), and participants found the game respectful and the painter metaphor easy to follow, although stroke resets, arm calibration and camera privacy still need work.

1 Introduction

People recovering from a stroke or an injury are usually sent home with a list of arm exercises to repeat every day. The exercises are dull, nobody is watching, and many people simply stop. When we surveyed existing rehabilitation games in Assignment 1, most of them relied on cartoon graphics and child-like rewards, which older players can reasonably find patronising. In Assignment 2 we proposed *vocational role-play* as an alternative: if each movement is part of familiar adult work, it has a purpose beyond the count. For this assignment we built a **vertical slice** of that idea. One trade, the House Painter, works end to end, from a therapist's reference movement to the player's pay slip. The other trades appear in the menu but cannot be played yet.

2 Conceptual Design

2.1 Metaphor

Our conceptual model is *a working day as a house painter* (Table 1). The point is that players should not have to learn a new set of rules; they can lean on a mental model they already have [1]. Most people know that tape marks where the paint should go, that a careful painter stays inside it, and that a working day has a start, a lunch break and an end. In Assignment 2 the trade was gardening. We switched to painting because a brush stroke maps naturally onto a continuous arm path, and because a wall slowly filling with colour makes progress visible at a glance.

Game element	Stands for	Design intent
Masking tape	Exercise path	Familiar "paint here" signifier; tape width is the tolerance
Brush	User's hand	Direct mapping: the brush moves as the hand moves
Paint appearing	Being on the path	Continuous feedback without numbers
Stroke / wall / job	Repetition / set / session	Natural hierarchy: "two more strokes on this wall"
Clock 9:00–17:00, lunch	Session progress, rest	Moves with work done, not real time, so there is no time pressure
Wage and pay slip	Effort and summary	Adult motivation; never deducted; clinical numbers kept separate

Table 1: Conceptual model.

2.2 Design principles

Table 2 shows where Norman's principles [1] appear in the interface. We also designed around Fitts' law [2], keeping targets large and close to the hand that has to reach them, and around Hick's law, by never showing more than six choices on one screen. Contrast follows WCAG 2.2 [3]. Finally, we treated *dignity* as a design requirement in its own right: the wording is adult, there is no childish praise, and nothing is ever taken away as a penalty.

Changes from the low-fidelity design. Several things changed after the low-fidelity prototype. Gardening became painting. The two photos of the arm that we had planned to use for sizing were replaced by a live three-second measurement. Selecting by making a fist became dwell selection, since grip strength is often exactly what the player is trying to recover. The clock no longer runs in real time from 9 to 5 but moves forward as work gets done, and seated play became the default.

3 System Design and Architecture

Principle	In the prototype
Discoverability	Few large, labelled buttons; Help and Back on every screen; “Raise your hand to point” hint
Feedback	Paint under the brush, glowing tape, filling dwell ring, sounds, spoken and captioned messages
Conceptual model	Painter, shift and wage metaphor; each wall opens with an animated demo of the stroke
Affordances	Raised, bordered buttons at least 72 px high; the brush drawn as a round tool head
Signifiers	Dashed tape with direction arrows, green start circle, dashed end circle
Mapping	Mirrored camera image; tape anchored to the user’s own shoulder; clock fills left to right
Constraints	Paint only inside the tape; a stroke must begin on the start circle; “End shift” asks to confirm

Table 2: Norman’s design principles in the interface.

3.1 Architecture and technology

Day Shift is a single-page web app with no server behind it (Figure 1). Everything, including pose estimation and video analysis, runs in the browser. As a result, camera images never leave the player’s computer, and the whole app can be hosted as static files. Table 3 lists the stack.

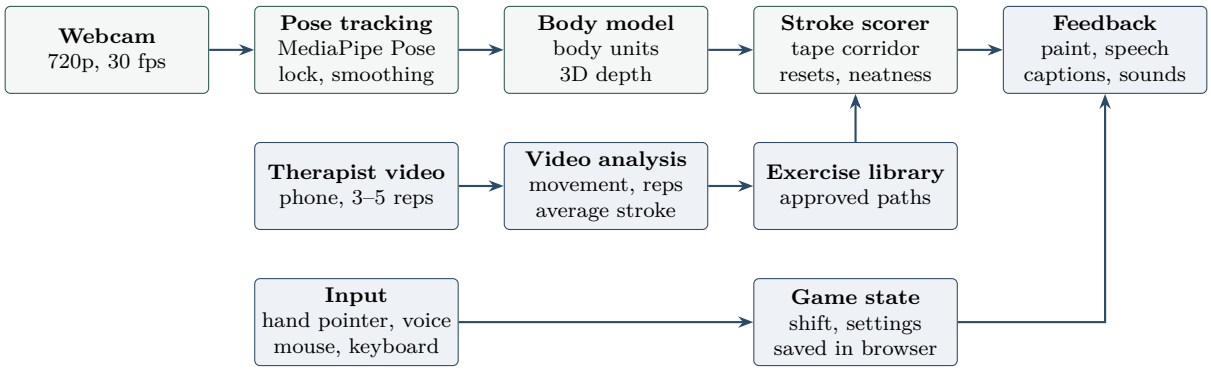


Figure 1: Architecture. Top: the live loop that runs for every camera frame. Middle: turning a therapist’s video into an exercise, which reuses pose tracking and the body model. Bottom: menus and saved state.

Concern	Technology	Why
Application	React 18, TypeScript, Vite	Component screens; type-checked geometry code
Pose	MediaPipe PoseLandmarker (BlazePose), full or lite model, bundled	Free, GPU in the browser, gives 3D world landmarks
State	zustand, saved to localStorage	Settings, progress and an unfinished shift survive a reload
Speech	Web Speech API: synthesis (out), recognition (in)	No downloads or keys; English (India) and Hindi voices
Sound, motion	Web Audio tones; Framer Motion	Full control of pitch and volume; motion can be turned off
Type, icons	Atkinson Hyperlegible [9]; Lucide icons	Designed for low vision; every icon has a label

Table 3: Technology stack.

3.2 Tracking the user

MediaPipe PoseLandmarker [4] gives us 33 body landmarks per frame, both as image positions and as rough 3D “world” coordinates. The tracker can detect up to three people, so we use a **target lock** that follows the torso closest to the player’s last known position; someone walking past in the background is ignored. Raw landmarks jitter, so we smooth them with a **One-Euro filter** [5]. It smooths heavily when the hand is still and lightly when the hand moves fast, which keeps the brush steady without making it lag. On our laptop, pose inference took a median of **44 ms** per frame with the full model, and a lighter model is available for slower machines. For voice commands we act on the recogniser’s first partial result instead of waiting for the final one, which took 1–3 s during formative testing. Spoken feedback uses the computer’s built-in voices and starts immediately.

3.3 Fitting one exercise to every person

Arm length. At the start, the player holds their arm out to the side for three seconds. We take upper-arm length as the shoulder-to-elbow distance in the image and forearm length as the elbow-to-wrist distance, and

add the two to get arm length L. A useful property of projection is that a bone can only *look* shorter than it really is (when it turns toward the camera), never longer. So instead of the median, we use a high percentile (the 85th) of each length. If the arm is consistently seen to be longer during play, the estimate is raised.

Body units. Exercise paths are stored relative to the joint that moves, and measured in the player's own segment lengths. For the arm, the origin O is the shoulder, x points outward along the shoulder line (mirrored for the left arm) and y points up. A wrist position P is stored and replayed as

$$p = \frac{R\,(P - O)}{L}, \qquad P = O_{\text{user}} + L_{\text{user}}\,R^{\top} p.$$

In these units the point $(1, 0)$ always means "the arm fully stretched out sideways at shoulder height", whoever the player is. If a therapist's wrist ends at 0.92 arm lengths, the tape ends at 0.92 of *each* player's own arm length. The wrist and shrug exercises work the same way, using hand length and shoulder height instead.

Distance. Players rarely sit perfectly still. Between strokes we compare the current shoulder width with the calibrated one and slowly rescale all lengths to match. During a stroke the scale is frozen, so the tape never shifts under the brush. If the player's size in the image stays outside 70–145% of the calibrated size for 1.5 s, the game pauses and asks them to move back to their spot.

3.4 Depth from a single webcam

A front-facing camera under-reports movements that go toward it, such as forward raises and elbow bends. A 60° forward raise, for example, looks like 0° from the front. Because the camera is far away compared with how far the arm moves in depth, we can treat the projection as roughly orthographic. A segment of calibrated length L_0 that appears with length l must then have a depth component

$$\Delta z = \sqrt{\max\,(0,\; L_0^2 - l^2)},$$

and we take its sign (toward or away from the camera) from MediaPipe's world coordinates. Chaining shoulder, elbow and wrist gives 3D joint angles. In our unit tests, a 60° forward raise reads as 60° in 3D and 0° in 2D. When a bone points almost straight at the camera the estimate becomes unstable, and we mark it as unreliable. Depth also gives the player a visual cue, since the brush gets bigger as the hand comes forward. We considered depth cameras and monocular depth networks, but people do not have the first at home, and the second were too slow in a browser.

3.5 From a therapist's video to an exercise

In Expert mode, a therapist records a front-facing phone video of 3–5 slow repetitions and uploads it. No photos or body measurements are needed. The app then:

1. runs pose tracking on the video at 15 frames per second and picks whichever arm moves more;
2. calibrates from the video itself, since the arm passes flat to the camera at some point;
3. converts every frame into body units and 3D angles;
4. decides what the main movement is with a simple rule (shoulder lift, elbow bend or sideways swing, whichever varies most), without any additional AI model;
5. cuts the movement into repetitions, drops outliers, and averages the rest into a single path.

The therapist then sees the tracked path drawn over the video, along with an angle chart. They can leave out bad repetitions, set the tape width and the number of strokes, name the exercise and approve it.

3.6 Scoring a stroke

A stroke begins when the brush rests on the start circle for 0.6 s. From then on, the brush position is projected onto the tape and progress can only move forward. Paint is only laid inside the tape, and the stroke is complete at 95% progress (plus an end hold, if the exercise has one). When something goes wrong, only the current stroke restarts, never the whole wall (Table 4). We score **neatness** from 0 to 100 by comparing the performed path with the reference path using Dynamic Time Warping [6]. The player earns Rs 40 per stroke, a neatness bonus of up to Rs 20, and Rs 50 for each finished wall. **Pay** is never deducted.

4 Prototype Description

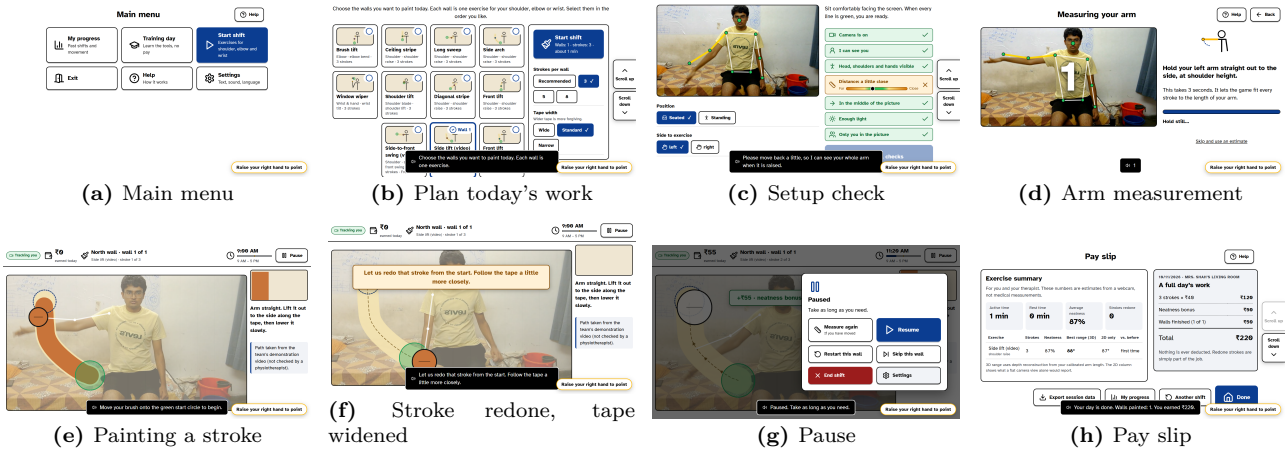
4.1 User journey

The whole flow works end to end: **Main menu** → **Plan today's work** → **Setup check** → **Arm measurement** → **Training (first time)** → **Shift** → **Pay slip** → **My progress** (Figure 2). In *Plan today's work*, the player (or a carer) picks which walls to paint, how many strokes to do and how wide the tape should be,

Situation	Rule	Response
Brush off the tape	more than 1 s	Stroke restarts with a calm spoken line
Leaning the trunk sideways	more than 15°	“Keep your back straight”; stroke restarts
Hiking the shoulder	small / large	Spoken hint first; only a large hike restarts
User or wrist not visible	tracking lost	Auto-pause; never counted as the user’s error
Repeated misses	2 misses / 3 neat strokes	Tape widens by 20% / narrows back

Table 4: Error rules. All thresholds were loosened after formative testing (Section 6.1).

choosing from eleven shoulder, elbow and wrist exercises. The *setup check* then goes through a short list, one line at a time: is the person visible, at a good distance, centred, well lit, and alone in the frame? During the *shift*, the screen shows the wall with the tape and the brush, with a faint camera image behind them. After each stroke a chime plays and the earnings appear. *Pausing* (with a button, the Space key, a voice command, or automatically when tracking is lost) offers Resume, Measure again, Skip this wall and End shift. On the *pay slip*, the wage is kept apart from a labelled movement summary of strokes, neatness and range of motion. Progress is saved after every stroke, so nothing is lost if the browser closes.



(a) Main menu **(b)** Plan today’s work **(c)** Setup check **(d)** Arm measurement

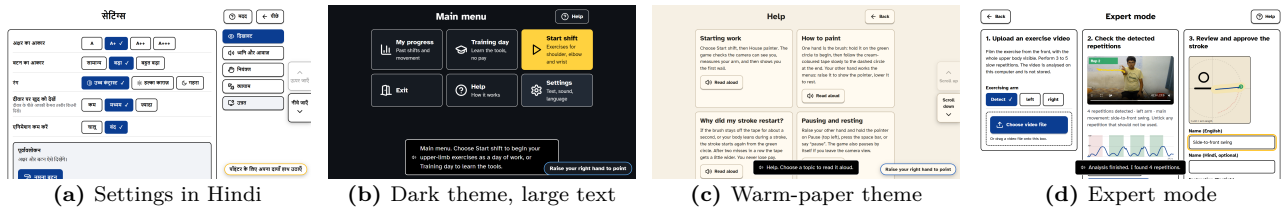
(e) Painting a stroke **(f)** Stroke redone, tape widened **(g)** Pause **(h)** Pay slip

Figure 2: The Painter journey, captured from the running prototype.

4.2 Interaction design for older adults

Many of our design decisions follow from age-related changes in vision, hearing and motor control. Table 5 summarises the main features and the reasoning behind each one.

Mouse, touch and keyboard (Space to pause, R to repeat, Esc to go back) work on every screen as fallbacks. All settings apply immediately and are saved (Figure 3). They cover text and button size, theme, language, voice and sound, captions, hand control, dwell time, the exercising arm, seated or standing play, tape width, posture checks, shift length and reduced motion.



(a) Settings in Hindi **(b)** Dark theme, large text **(c)** Warm-paper theme **(d)** Expert mode

Figure 3: Personalisation and the therapist’s view.

5 Compromises in Prototyping

With limited time, we had to choose where to go deep and where to stay shallow. Table 6 summarises those choices.

6 User Evaluation

Feature	Design	Reason
Typeface	Atkinson Hyperlegible, 24 px by default (20–34 px)	Built to tell similar letters apart for low vision
Colour	High contrast (default, 21:1), warm paper, dark theme	Exceeds WCAG AAA; warm theme reduces glare
Redundancy	Every state shown with icon, text and colour, and spoken	Works with colour blindness and hearing loss
Targets	Buttons 72–108 px high, ≤6 per screen	Fitts' and Hick's laws; tremor tolerance
Hand pointer	The *free* hand points; the exercising hand paints	Menus never interrupt the exercise
Raise to point	Pointer appears only when the hand is raised	A hand resting on the lap never clicks
Dwell select	Hover fills a ring in 1.5 s (1–3 s)	No pinch or grip, which is often impaired
Reach box	Small box on the hand's own side maps to the whole screen	Small movements; no reaching across the body
Side layout	Main buttons mirror to the pointing hand's side	Shorter movements; large scroll buttons there too
Voice commands	pause, resume, repeat, next, back, help (English, Hindi)	Hands-free control; optional
Speech, captions	Every instruction spoken (rate 0.9, adjustable) and captioned	Low vision, low literacy, muted laptops
Earcons	Soft tones of 260–1100 Hz, under 0.35 s, one meaning each	Age-related hearing loss affects high pitches
Language	Full English and Hindi: text, speech and voice commands	Target users in India
Tone	"Let us redo that stroke", never "wrong"; no confetti	Dignity and adult framing

Table 5: HCI features of the prototype.

6.1 Formative testing

Before running the study, we tested the prototype on ourselves, and with scripted runs in which recorded exercise videos were fed to the browser in place of the webcam. Table 7 lists the main problems we found and what we changed in response. We also wrote 31 automated tests on synthetic skeletons, covering calibration, 3D angles, scaling, stroke resets and voice matching.

6.2 Study design

Method. We ran a task-based usability test. Each participant played one supervised session in a controlled setting: a quiet room, a laptop webcam, and the participant sitting or standing 1.5–2 m from the camera. Everyone worked through the same seven tasks: (T1) get ready (setup check); (T2) arm measurement; (T3) training; (T4) paint one wall; (T5) take a break and continue; (T6) make the text bigger; and (T7) read the pay slip. Afterwards they filled in a Google Form with a Single Ease Question (SEQ, 1–7) for each task, the System Usability Scale (SUS) [7], five-point Likert items on the conceptual model, the interface and dignity, Borg CR10 effort ratings [8] from before and after the session, their preferred input method, and a few open-ended questions. SUS was scored in the standard way (0–100). For Likert items we report the mean and the share of participants who agreed (a 4 or 5), and we grouped the written comments into themes.

Participants. Nine people took part, coded P1–P9. Two were older adults aged 60–69 (P5, P6). The other seven were proxy users, five aged 18–29 and two aged 45–59. Seven played in English, one in Hindi, and one switched between the two. Six played seated, two standing, and one tried both. Self-rated comfort with technology averaged 4.2 ± 1.0 out of 5, with the two older adults rating themselves 3 and 4. Two participants had used a rehabilitation app before, and no physiotherapist took part.

6.3 Results

Usability. The mean SUS score was 73.3 ± 16.1 (Table 8). This is above the commonly used benchmark of 68 and falls in the "good" range, and six of the nine participants scored above 68 (Figure 4). With only nine people, though, the difference from 68 is not statistically significant ($t(8) = 1.00$, $p = .35$), so we read it as encouraging rather than conclusive. Scores were spread out, from 40 (P2) to 97.5 (P7), but most fell between 67.5 and 82.5. The two older adults scored 82.5 (P5) and 67.5 (P6), and the proxy users averaged 72.9 ± 18.0. P2 was a clear negative case: along with the lowest SUS score, they gave low ratings to both the wage and the brush feedback. The weakest SUS items were confidence in using the game (item 9, 3.7/5) and the belief that most people would learn it quickly (item 7, 3.8/5).

Task ease. Figure 5 shows each participant's rating for every task, so the spread is visible alongside the mean.

Dimension	Decision and justification
Vertical (deep)	The Painter trade end to end with real computer-vision scoring, all input and output modes, settings, two languages and saving. We chose depth over breadth because the main risk was whether webcam tracking could be fair and personal.
Horizontal (shallow)	Gardener, Postal sorter and Carpenter are shown as "coming later"; Help is static; no accounts, reminders, therapist dashboard or cloud sync.
Exercise content	Built-in paths follow standard exercise descriptions and team recordings; no physiotherapist has checked them yet. Expert mode lets a professional replace them.
Depth	Geometric estimate that assumes the user stays in place; approximate, not clinical-grade.
Movements	Shoulder, elbow and wrist. Finger and grip exercises need a hand-tracking model.
Voice input	Uses the browser's online speech service (Chrome and Edge); off by default.
Platform	Desktop browser with a webcam; one user per device; no clinical validation.
Wizard of Oz	None: everything in the demo is implemented.

Table 6: Prototyping compromises.

Finding	Change
Using the exercising hand for menus broke both	Free-hand pointer with raise-to-point
Small shifts in the chair caused resets	Adaptive scale; posture judged per stroke; 1 s grace; tape widening
A fully stretched arm went past the end of the tape	High-percentile arm length, refined during play
Voice commands felt slow or did not respond	Act on partial results; ignore the game's own voice
Scrolling was impossible and reaching across the body tiring	Scroll buttons and layout on the hand's side
Users could not choose their exercises	"Plan today's work" and "Skip this wall"
The hand pointer worked only after a shift began	Camera starts on the main menu

Table 7: Formative findings and the design changes they led to.

All tasks except T6 averaged 5.7 or higher: the setup check scored 6.2, training 6.3, painting a wall 5.8, and reading the pay slip 6.2. Two tasks deserve a closer look. Ratings for *arm measurement* (T2, mean 5.7) ranged all the way from 3 to 7, and holding the arm out for three seconds was clearly hard for some people. *Making the text bigger* (T6, 4.6) was the weakest task. Three of the seven people who rated it gave a 3 or 4, and two did not complete it at all. Our reading is that the settings are hard to find once a shift is under way.

Conceptual model. The painter metaphor was easy to pick up. Eighty-nine percent agreed that it made the exercises easy to understand, and the same share agreed that the masking tape showed them where to move. Everyone agreed that the arrows and the start circle showed how to begin a stroke, and that the animated demo showed what to do (Figure 6, top group). The wage was the weakest part of the metaphor: only 56% agreed that earning a wage made them want to keep going (mean 3.3). This is interesting, because 78% agreed that the pay slip kept pay separate from the movement summary, and 67% felt that exercising this way was more purposeful than plain repetitions. One possible reading is that the work framing gives the movements meaning, while the money itself adds little extra motivation, at least within a single short session.

Interface for older adults. The visual design was rated highly, with large text at 4.7, buttons at 4.7 and contrast at 4.3. Among the feedback channels, captions (4.8, with everyone agreeing) and spoken instructions (4.4) were the most appreciated. Hand control scored lower than the rest of the interface: raise-to-point received 3.6 and dwell selection 3.7 ($n = 7$, since two participants never used the hand pointer). The clearest problem was feedback after a reset. Only 44% understood why a stroke had restarted (mean 3.4), the lowest score among the feedback items, and tracking reliability was middling at 3.8.

Dignity and acceptance. This is where the results were strongest. Every participant agreed that the wording and visuals were respectful and never childish (4.8). Seventy-eight percent felt treated as an adult and did not feel judged when a stroke had to be redone, and 89% said they would not be embarrassed to play in front of family. Privacy was the main reservation. Only 56% were not worried while the camera was on (mean 3.3), and the two who disagreed made it the item with the most disagreement in Figure 6. Acceptance was still high. Likelihood of using the game at home or recommending it averaged 8.2 out of 10, and no one went below 7. Paint feedback on the tape was named as a useful feature most often (7 of 9), followed by the painter and wage idea, the free-hand pointer, and pause and rest breaks (6 each). The gardener was the most requested next trade (4 of 9).

Effort and comfort. Mean Borg effort was 3.9 before the session and 4.3 after it. Two proxy users already

Measure	Result	Note
SUS (0–100)	73.3 ± 16.1	Median 75, range 40–97.5, 95% CI 61.0–85.7; 6 of 9 above 68
SEQ, all tasks (1–7)	5.8	Lowest: T6 Make text bigger, 4.6 ± 1.3
Dignity, 3 items (1–5)	4.41 ± 0.74	Treated as an adult, respectful wording, not judged
Borg CR10 before / after	3.9/4.3	Change $+0.4 \pm 3.0$, Wilcoxon $p = .70$
Likely to use or recommend (0–10)	8.2 ± 1.0	All responses 7–10; 3 promoters, 6 passives, 0 detractors
Preferred input	Dwell 5, mouse/touch 3, voice 1	Both older adults preferred mouse/touch
Pace / difficulty / shift length	Pace "about right" 9/9; difficulty "about right" 5, "too easy" 4; length "about right" 6, "too short" 3	

Table 8: Summary of the user study (mean ± SD, $n = 9$ unless stated).

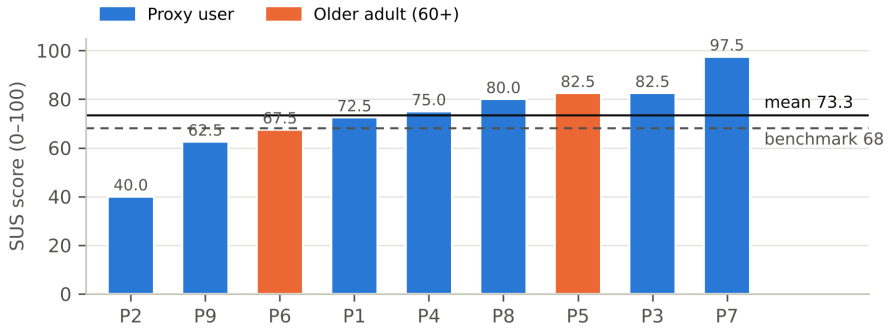


Figure 4: SUS score per participant, sorted from lowest to highest. Blue bars are proxy users and orange bars are older adults. The solid line is the mean (73.3) and the dashed line is the commonly used benchmark of 68.

rated their effort as high before starting (7 and 8), which makes the before–after comparison hard to interpret for them. Four participants rated their effort at 5 or more after the session. One of them was P6, an older adult, who went from 4 to 6 and reported arm fatigue. Two others reported mild discomfort, and one of these two also mentioned eye strain and feeling rushed. Nobody reported pain or dizziness, and everyone said the pace was about right. Since we only collected Borg ratings before and after the session, these data cannot tell us whether anyone crossed the stop threshold (Borg ≥ 5) during play.

Qualitative feedback. Five participants left written comments, and these fell into five themes. First, dwell selection felt slow, and keeping the hand still in the air was tiring while seated (P3, P4); P3 found voice commands easier. Second, the posture warning appeared and disappeared too quickly to explain why a stroke had restarted (P5), which fits the low rating on that item. Third, holding the arm up during calibration was difficult with a weak shoulder (P6). Fourth, having the camera on raised privacy concerns, because nothing on screen explained what happens to the video (P6). Fifth, participants wanted more variety and control: an adjustable dwell time, narrower tape, a choice of paint colours and rooms, a spoken countdown before each stroke, and keyboard shortcuts shown on screen (P3, P4, P7). There were also a few concrete suggestions, such as letting the elbow rest on an armrest during calibration (P6), prompting a rest after repeated shoulder hiking (P5), and adding strokes that go both up and down (P9).

Expert items. Six participants answered the optional questions about the exercises, but none of them were clinicians. They rated the exercises as appropriate for older adults (4.7) and were least convinced that the error rules made clinical sense (3.8). Without a professional among them, we treat these numbers only as rough indications.

6.4 Design changes from the evaluation

Table 9 ranks the main findings by severity and lists the change each one points to.

Limitations. This study has clear limits. The sample was small ($n = 9$), and only two participants belonged to the target group. The other seven were proxy users and no physiotherapist took part, so the results speak to usability rather than clinical value. Each participant played a single short session, so we cannot yet say anything about long-term use or adherence. We also did not record task completion, time on task, error counts, the goniometer comparison of 3D and 2D angles, or voice-command delay in this round. These are left for the next study.

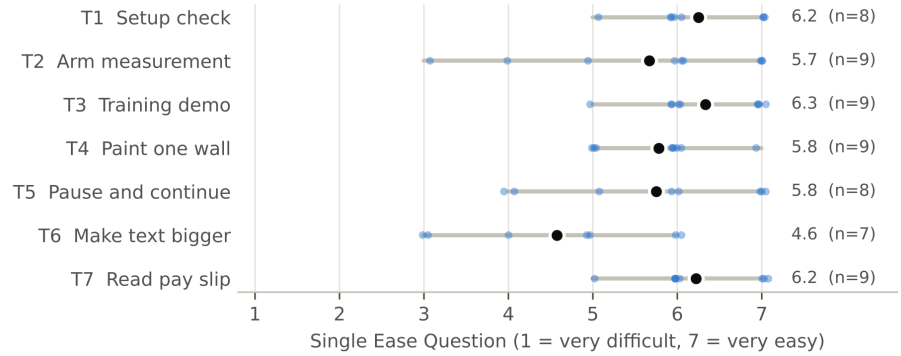


Figure 5: Single Ease Question per task (1 = very difficult, 7 = very easy). Each blue dot is one participant, the grey line spans the lowest to highest rating, and the black dot is the mean. The number on the right is the mean and how many participants rated the task; "Did not do" answers are excluded.

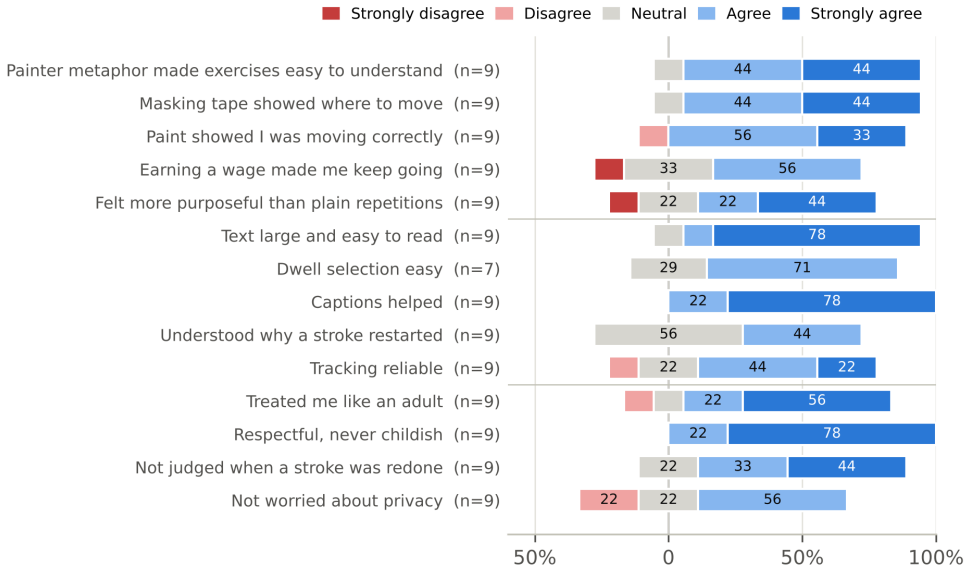


Figure 6: Selected Likert items (1–5) as percentages of respondents. Bars are centred on the middle of "Neutral", so agreement extends to the right and disagreement to the left. The three groups, top to bottom, are the conceptual model, the interface and dignity. Numbers inside bars are percentages; n is shown for each item because "Did not use" answers are excluded.

7 Ethics and Risk Assessment

Table 10 lists the main risks we identified and how we addressed each of them.

Consent. We collected consent at the start of the Google Form. The form was in English, and a team member translated it into Hindi when needed. Before continuing, participants had to agree to the following statement: *"I have read the information above (or it was read to me). I understand that participation is voluntary, that I can stop at any time, and that no sensitive personal or health data will be collected."* Anyone who did not agree was taken straight to the end of the form.

Four participants reported post-session effort of 5 or more on the Borg scale, although none reported pain or dizziness. Because effort was only rated before and after the session, the stop rule could not be applied during play. In future sessions we will ask for a rating between walls.

8 Conclusion

This prototype shows that the idea from Assignment 2 can work on ordinary hardware. A single webcam and in-browser pose tracking are enough to fit one reference movement to players of different sizes, to recover movement toward the camera, and to score strokes in a forgiving way. A therapist can turn a phone video into a playable exercise without taking measurements or relying on additional models. Around this technical core, the interface is designed with ageing users in mind: readable type and strong contrast, large dwell targets for the free hand, spoken and captioned feedback, voice and keyboard alternatives, Hindi, and an adult metaphor

Severity	Finding (evidence)	Change
Major	Stroke resets not understood (3.4/5; P5)	Spoken posture cue ("sit up straight") with its own tone; warning stays until the next stroke starts
Major	Arm measurement tiring (T2 min 3; P6)	Supported calibration with the elbow on an armrest, or a shorter hold; offer a rest afterwards
Major	Privacy worry (3.3/5; P6)	Permanent "processed on this device, not recorded" badge beside the camera view
Minor	Text size hard to change (T6 4.6)	Text-size control on the pause menu and main menu; spoken hint "say *bigger text*"
Minor	Dwell slow or tiring (3.7/5; P3, P4)	Shorter default dwell in menus; make the existing 1–3 s setting easier to find; suggest voice after repeated slow selections
Minor	Effort not monitored during play (4 of 9 had post-session Borg ≥ 5)	Ask a Borg rating between walls and suggest a rest at 5 or above
Minor	Too easy for some proxy users (4/9)	Progressive tape narrowing; colour and room choices for variety

Table 9: Findings ranked by severity and the design changes they lead to.

Risk	**Mitigation**
Fatigue, pain, falls	Seated by default; mild exercises; short sessions; no time pressure; rests and lunch break; pause at any time; stop if pain, dizziness or Borg ≥ 5; a team member supervises; "Not a medical device" notice in the app
Privacy	Camera frames are processed on the device and never stored or sent; only an optional nickname, settings and numbers are saved; participants coded P1, P2, . . . ; no video or audio recorded
Voice data	Voice commands are off by default, and Settings explains that audio goes to the browser's speech service
Feeling judged	Adult wording; no deductions; participants told "we are testing the game, not you"
Clinical claims	Movement numbers labelled as estimates; no diagnosis or medical advice

Table 10: Risks and mitigations.

without penalties.

Our usability study with nine participants (two older adults and seven proxy users) broadly supports these choices. The game reached a mean SUS score of 73.3, and every task except changing the text size was rated as easy (SEQ of 5.7 or more out of 7). Participants understood the painter metaphor and the masking tape, and all of them found the game respectful rather than childish. On average they rated their likelihood of using or recommending it at 8.2 out of 10. The study also showed us what does not work yet. Participants often could not tell why a stroke had restarted, holding the arm up for calibration was tiring, the camera raised privacy concerns, and the wage motivated only about half of them.

The main limitations are the small sample with few older adults, the lack of validation by a physiotherapist, and the approximate nature of our depth estimate. Our next steps are the design changes in Table 9 (spoken posture cues, supported calibration and an on-screen privacy badge), followed by a larger study with older adults and a physiotherapist that includes the accuracy and latency measurements we could not run this time. Further ahead are hand tracking for finger exercises, more trades (starting with the gardener), and a dashboard for therapists.

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